MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, ADVANCED LOW POWER SCHOTTKY TTL, NOR GATES, MONOLITHIC SILICON

This specification is approved for use by the Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for monolithic silicon, advanced low power Schottky TTL, positive NOR logic gating microcircuits. Two product assurance classes and a choice of case outlines and lead finishes are provided and are reflected in the complete part number.

1.2 Part number. The part number shall be in accordance with MIL-M-38510.

1.2.1 Device type. The device type shall be as follows:

Device type	Circuit
01	Quadruple, 2-input positive NOR gate
02	Triple, 3-input positive NOR gate

1.2.2 Device class. The device class shall be the product assurance level as defined in MIL-M-38510.

1.2.3 Case outline. The case outline shall be designated as follows:

Outline letter	Case outline (MIL-M-38510, appendix C)
A	F-1 (14-lead, 1/4" x 1/4"), flat package
B	F-3 (14-lead, 3/16" x 1/4"), flat package
C	D-1 (14-lead, 1/4" x 3/4"), dual-in-line package
D	F-2 (14-lead, 1/4" x 3/8"), flat package
X	C-2 (20-terminal, .350" x .350"), square chip carrier package, option A
Y	C-2 (20-terminal, .350" x .350"), square chip carrier package, option B

1.3 Absolute maximum ratings.

Supply voltage range	-0.5 V dc to 7.0 V dc
Input voltage range	-1.5 V dc at -18 mA to 7.0 V dc
Storage temperature range	-65°C to +150°C
Maximum power dissipation (P_D) per device: 1/	
Device type 01	22 mW
Device type 02	21.9 mW
Lead temperature (soldering, 10 seconds)	300°C
Thermal resistance, junction to case (θ_{JC}):	
Cases A, B, D	70°C/W
Case C	50°C/W
Cases X, Y	60°C/W
Junction temperature (T_J)	175°C

1.4 Recommended operating conditions.

Supply voltage	4.5 V dc minimum to 5.5 V dc maximum
Minimum high level input voltage (V_{IH})	2.0 V dc
Maximum low level input voltage (V_{IL})	0.8 V dc
Case operating temperature range	-55°C to +125°C

1/ Must withstand the added P_D due to short circuit test (e.g. I_O).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Rome Air Development Center (RBE-2), Griffiss AFB, NY 13441, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-M-38510 - Microcircuits, General Specification for.

STANDARDS

MILITARY

MIL-STD-883 - Test Methods and Procedures for Microelectronics.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Detail specifications. The individual item requirements shall be in accordance with MIL-M-38510, and as specified herein.

3.2 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-M-38510, and herein.

3.2.1 Case outlines. The case outlines shall be as specified in 1.2.3.

3.2.2 Logic diagrams and terminal connections. The logic diagrams and terminal connections shall be as specified on figure 1.

3.2.3 Truth tables and logic equations. The truth tables and logic equations shall be as specified on figure 2.

3.2.4 Schematic circuits. The schematic circuits shall be submitted to the preparing activity prior to inclusion of a manufacturer's device in this specification and shall be submitted to the qualifying activity as a prerequisite for qualification. All manufacturers' schematics shall be maintained and available upon request.

3.3 Lead material and finish. The lead material and finish shall be in accordance with MIL-M-38510 (see 6.5).

3.4 Electrical performance characteristics. The electrical performance characteristics are as specified in table I, and apply over the full recommended case operating temperature range, unless otherwise specified.

3.5 Electrical test requirements. The electrical test requirements for each device class shall be the subgroups specified in table II. The electrical tests for each subgroup are described in table III.

3.6 Marking. Marking shall be in accordance with MIL-M-38510. At the option of the manufacturer, marking of the country of origin may be omitted from the body of the microcircuit, but shall be retained on the initial container.

3.7 Microcircuit group assignment. The devices covered by this specification shall be in microcircuit group number 8 (see MIL-M-38510, appendix E).

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TABLE I. Electrical performance characteristics.

Test	Symbol	Conditions $-55°C \le T_C \le +125°C$	Device type	Limits Min	Limits Max	Unit
High-level output voltage	V_{OH}	V_{CC} = 4.5 V, I_{OH} = -400 µA V_{IL} = 0.8 V	All	2.5		V
Low-level output voltage	V_{OL}	V_{CC} = 4.5 V, V_{IH} = 2.0 V, I_{OL} = 4.0 mA	All		0.4	V
Input clamp voltage	V_{IC}	V_{CC} = 4.5 V, I_{IN} = -18 mA T_C = +25°C	All		-1.5	V
Low-level input current	I_{IL}	V_{CC} = 5.5 V, V_{IN} = 0.4 V	All	0	-100	µA
High-level input current	I_{IH1}	V_{CC} = 5.5 V, V_{IN} = 2.7 V	All		20	µA
	I_{IH2}	V_{CC} = 5.5 V, V_{IN} = 7.0 V	All		100	µA
Output current 1/	I_O	V_{CC} = 5.5 V, V_O = 2.25 V	All	-30	-112	mA
Supply current, outputs high	I_{CCH}	V_{CC} = 5.5 V, V_{IN} = 0 V	01		2.2	mA
			02		1.8	mA
Supply current, outputs low	I_{CCL}	V_{CC} = 5.5 V, V_{IN} = 4.5 V	All		4.0	mA
Propagation delay time, low to high level	t_{PLH}	V_{CC} = 5.0 V, C_L = 50 pF, R_L = 500Ω	01	3	14	ns
			02	4	22	
Propagation delay time, high to low level	t_{PHL}	V_{CC} = 5.0 V, C_L = 50 pF, R_L = 500Ω	01	3	11	ns
			02	3	10	

1/ The output conditions have been chosen to produce a current that closely approximates one half of the true short-circuit output current, I_{OS}.

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TABLE II. Electrical test requirements.

MIL-STD-883 test requirements	Subgroups (see table III)	
	Class S devices	Class B devices
Interim electrical parameters (pre burn-in) (method 5004)	1	1
Final electrical test parameters (method 5005)	1*,2,3,9,10,11	1*,2,3,9
Group A test requirements (method 5005)	1,2,3,9,10,11	1,2,3,9
Group B test requirements (method 5005) subgroup 5	1,2,3,9,10,11	N/A
Group C end-point electrical parameters (method 5005)	N/A	1,2,3
Additional electrical subgroups for group C periodic inspections	N/A	10,11
Group D end-point electrical parameters (method 5005)	1,2,3	1,2,3

*PDA applies to subgroup 1 (see 4.2c.).

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with MIL-M-38510 and methods 5005 and 5007, as applicable, of MIL-STD-883, except as modified herein.

4.2 Screening. Screening shall be in accordance with method 5004 of MIL-STD-883, and shall be conducted on all devices prior to qualification and quality conformance inspection. The following additional criteria shall apply:

a. Burn-in (method 1015 of MIL-STD-883).

(1) Test condition D or E, using the circuit shown on figure 3, or equivalent.

(2) T_A = +125°C minimum.

b. Interim and final electrical test parameters shall be as specified in table II, except interim electrical parameters test prior to burn-in is optional at the discretion of the manufacturer.

c. The PDA for class S devices shall be as specified in MIL-M-38510. The PDA for class B devices shall be 10 percent based on failures from group A, subgroup 1 test after cooldown as final electrical test in accordance with method 5004 of MIL-STD-883, and with no intervening electrical measurements. If interim electrical parameter tests are performed prior to burn-in, failures resulting from pre burn-in screening may be excluded from the PDA. If interim electrical parameter tests prior to burn-in are omitted, then all screening failures shall be included in the PDA. The verified failures of group A, subgroup 1, after burn-in divided by the total number of devices submitted for burn-in in that lot shall be used to determine the percent defective for that lot, and the lot shall be accepted or rejected based on the PDA for the applicable device class.

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3 Qualification inspection. Qualification inspection shall be in accordance w. . MIL-M-38510. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

4.4.1 Group A inspection. Group A inspection shall be in accordance with table I of method 5005 of MIL-STD-883 and as follows:

a. Tests shall be as specified in table II herein.

b. Subgroups 4, 5, 6, 7, and 8 shall be omitted.

4.4.2 Group B inspection. Group B inspection shall be in accordance with table II of method 5005 of MIL-STD-883. Electrical parameters shall be as specified in table II herein.

4.4.3 Group C inspection. Group C inspection shall be in accordance with table III of method 5005 of MIL-STD-883 and as follows:

a. End point electrical parameters shall be as specified in table II herein.

b. Subgroups 3 and 4 shall be added to the group C inspection requirements for class B devices and shall consist of the tests, conditions, and limits specified for subgroups 10 and 11 of group A.

c. Steady-state life test (method 1005 of MIL-STD-883) conditions:

(1) Test condition D or E, using the circuit shown on figure 3, or equivalent.

(2) T_A = +125°C minimum.

(3) Test duration: 1,000 hours, except as permitted by appendix B of MIL-M-38510.

4.4.4 Group D inspection. Group D inspection shall be in accordance with table IV of method 5005 of MIL-STD-883. End point electrical parameters shall be as specified in table II herein.

4.5 Methods of inspection. Methods of inspection shall be specified as follows:

4.5.1 Voltage and current. All voltages given are referenced to the microcircuit ground terminal. Currents given are conventional and positive when flowing into the referenced terminal.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-38510.

6. NOTES

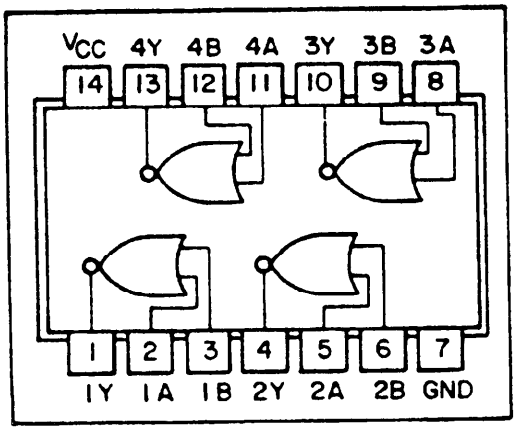
6.1 Notes. The notes specified in MIL-M-38510 are applicable to this specification.

6.2 Intended use. Microcircuits conforming to this specification are intended for ... nal equipment design applications and logistic support of existing equipment.

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Device type 01

CASES A, B, C AND D



Device type 02

CASES A, B, C AND D

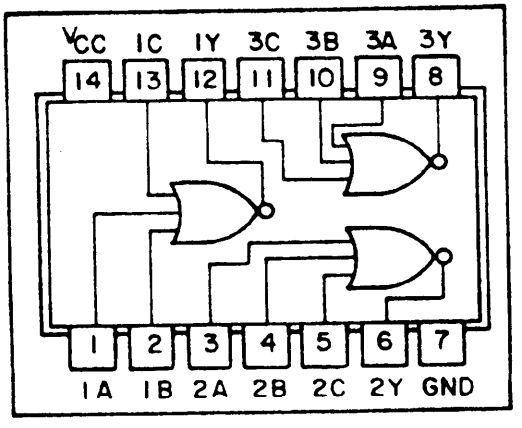
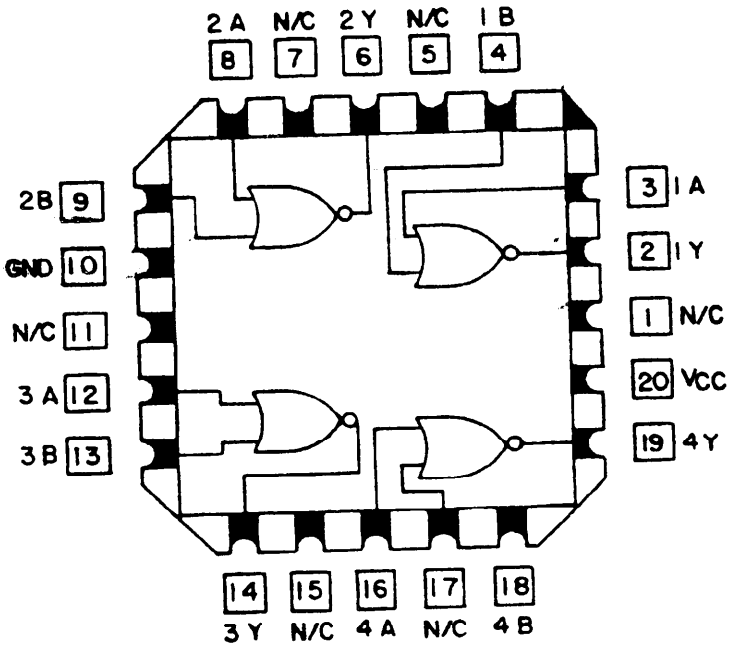


FIGURE 1. Logic diagrams and terminal connections (top view).

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Device type 01
Cases X,Y



Device type 02
Cases X,Y

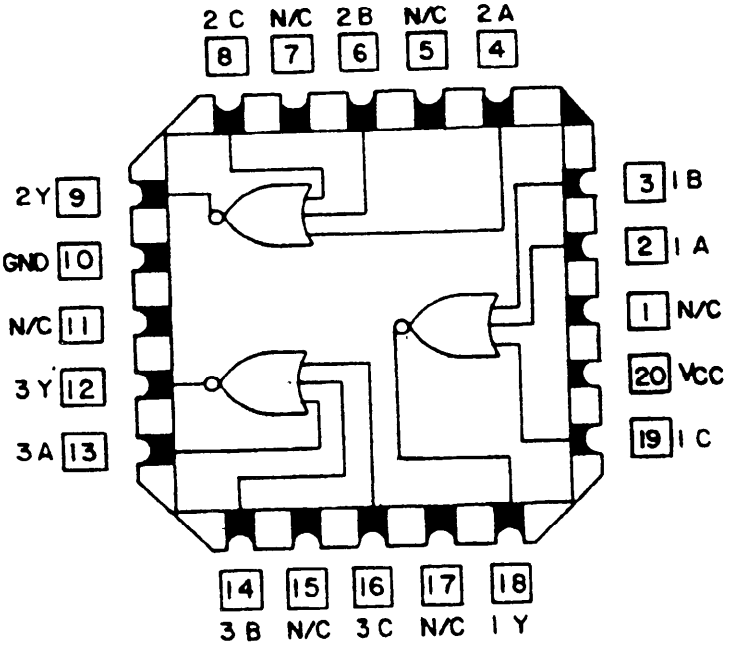


FIGURE 1. Logic diagrams and terminal connections - Continued.

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Device type 01

TRUTH TABLE (EACH GATE)	
INPUTS	OUTPUT
A B	Y
L L	H
L H	L
H L	L
H H	L

$Y = \overline{A + B}$

Device type 02

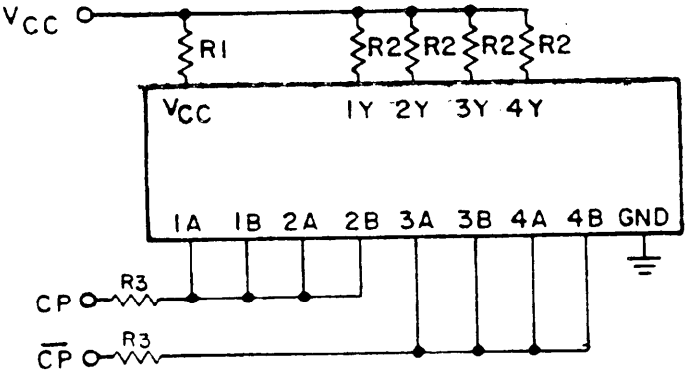
TRUTH TABLE (EACH GATE)			
INPUTS			OUTPUT
A	B	C	Y
L	L	L	H
H	L	L	L
L	H	L	L
L	L	H	L
H	H	L	L
H	L	H	L
L	H	H	L
H	H	H	L

$Y = \overline{A + B + C}$

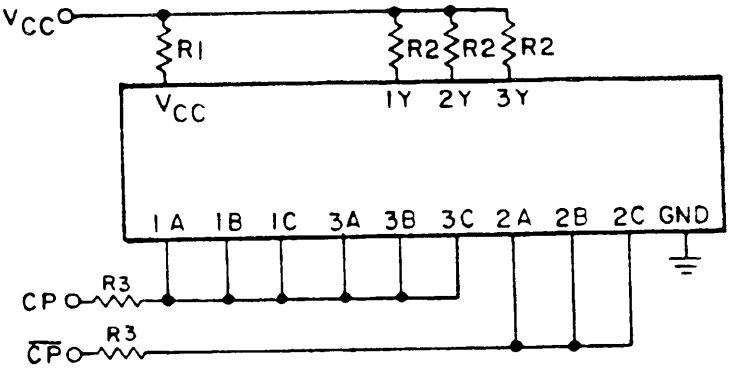
FIGURE 2. Truth tables and logic equations.

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Device type 01



Device type 02

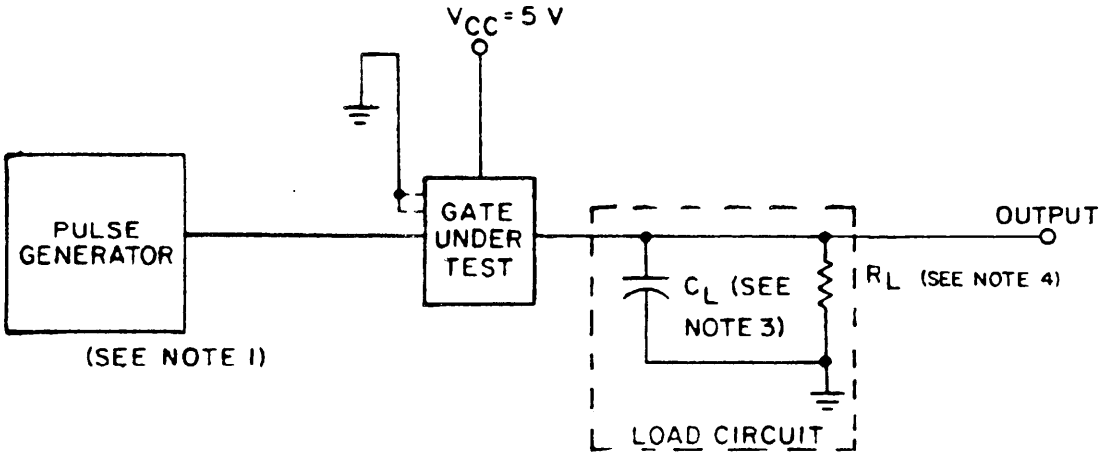


NOTES:
1. CP or $\overline{CP}$ = 100 kHz ±50% square wave; duty cycle = 50 ±15%; V_{IL} = -0.5 V minimum to 0.8 V maximum; V_{IH} = 2.0 V minimum to 5.5 V maximum.
2. R1 = 10Ω maximum; R2 = 1k ±5%; R3 = 27Ω maximum.
3. V_{CC} shall be high enough to insure 5.0 V minimum is present at device V_{CC} terminal.

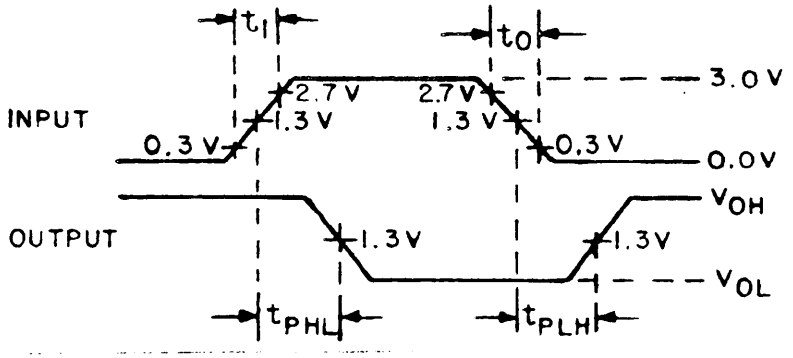
FIGURE 3. Burn-in and life test circuits.

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Device types 01 and 02



TEST CIRCUIT



NOTES:
1. Input pulse has the following characteristics: $t_1 = t_0 = 6 \pm 1.5$ ns, PRR = 1 MHz, $Z_{OUT} \cong 50\Omega$.
2. Inputs not under test are at ground.
3. C_L = 50 pF ±10%, including scope probe, wiring, and stray capacitance without package in test fixture.
4. $R_L = 499\Omega \pm 1\%$.
5. Voltage measurements are to be made with respect to network ground terminal.

FIGURE 4. Switching time test circuit and waveforms for device types 01 and 02.

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TABLE III. Group A inspection for device type 01.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases X, Y 1/	2	3	4	6	8	9	10	12	13	14	16	18	19	20	Measured terminal	Limits		Unit
			Cases A,B,C,D	1	2	3	4	5	6	7	8	9	10	11	12	13	14				
			Test no.	1Y	1A	1B	2Y	2A	2B	GND	3A	3B	3Y	4A	4B	4Y	V_{CC}		Min	Max	
1 T_C = 25°C	V_{OH}	3006	1	-0.4 mA	0.8 V	0.8 V				GND							4.5 V	1Y	2.5		V
		"	2				-0.4 mA	0.8 V	0.8 V	"							"	2Y	"		"
		"	3							"	0.8 V	0.8 V	-0.4 mA				"	3Y	"		"
		"	4							"				0.8 V	0.8 V	-0.4 mA	"	4Y	"		"
	V_{OL}	3007	5	4.0 mA	2.0 V	GND				"							"	1Y		0.4	"
		"	6	4.0 mA	GND	2.0 V				"							"	1Y		"	"
		"	7				4.0 mA	2.0 V	GND	"							"	2Y		"	"
		"	8				4.0 mA	GND	2.0 V	"							"	2Y		"	"
		"	9							"	2.0 V	GND	4.0 mA				"	3Y		"	"
		"	10							"	GND	2.0 V	4.0 mA				"	3Y		"	"
		"	11							"				2.0 V	GND	4.0 mA	"	4Y		"	"
		"	12							"				GND	2.0 V	4.0 mA	"	4Y		"	"
	V_{IC}		13		-18 mA	-18 mA				"							"	1A		-1.5	"
			14					-18 mA	-18 mA	"							"	1B		"	"
			15							"	-18 mA	-18 mA					"	2A		"	"
			16							"				-18 mA	-18 mA		"	2B		"	"
			17							"							"	3A		"	"
			18							"							"	3B		"	"
			19							"							"	4A		"	"
			20							"							"	4B		"	"
	I_{IL}	3009	21		0.4 V	0.4 V				"							5.5 V	1A	2/	2/	μA
		"	22					0.4 V	0.4 V	"							"	1B	"	"	"
		"	23							"	0.4 V	0.4 V					"	2A	"	"	"
		"	24							"				0.4 V	0.4 V		"	2B	"	"	"
		"	25							"							"	3A	"	"	"
		"	26							"							"	3B	"	"	"
		"	27							"							"	4A	"	"	"
		"	28							"							"	4B	"	"	"
	I_{IH1}	3010	29		2.7 V	2.7 V				"							"	1A		20	μA
		"	30					2.7 V	2.7 V	"							"	1B		"	"
		"	31							"	2.7 V	2.7 V					"	2A		"	"
		"	32							"				2.7 V	2.7 V		"	2B		"	"
		"	33							"							"	3A		"	"
		"	34							"							"	3B		"	"
		"	35							"							"	4A		"	"
		"	36							"							"	4B		"	"
	I_{IH2}	"	37		7.0 V	7.0 V				"							"	1A		100	μA
		"	38					7.0 V	7.0 V	"							"	1B		"	"
		"	39							"	7.0 V	7.0 V					"	2A		"	"
		"	40							"				7.0 V	7.0 V		"	2B		"	"
		"	41							"							"	3A		"	"
		"	42							"							"	3B		"	"
		"	43							"							"	4A		"	"
		"	44							"							"	4B		"	"
	I_O 3/	3011	45	2.25 V	GND	GND				"							"	1Y	-30 4/	-115 4/	mA
		"	46				2.25 V	GND	GND	"							"	2Y	"	"	"
		"	47							"	GND	GND	2.25 V				"	3Y	"	"	"
		"	48							"				GND	GND	2.25 V	"	4Y	"	"	"
	I_{CCH}	3005	49		GND	GND		GND	GND	"	GND	GND		GND	GND		"	V_{CC}		2.2	"
	I_{CCL}	3005	50		4.5 V	4.5 V		4.5 V	4.5 V	"	4.5 V	4.5 V		4.5 V	4.5 V		"	V_{CC}		4.0	"

See footnotes at end of table.

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TABLE III. Group A inspection for device type 01. - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases X, Y 1/	2	3	4	6	8	9	10	12	13	14	16	18	19	20	Measured terminal	Limits		Unit
			Cases A,B,C,D	1	2	3	4	5	6	7	8	9	10	11	12	13	14		Min	Max	
			Test no.	1Y	1A	1B	2Y	2A	2B	GND	3A	3B	3Y	4A	4B	4Y	V_{CC}				
2	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = +125°C and V_{IC} tests are omitted.																				
3	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																				
9 T_C = 25°C	t_{PLH}	3003 Fig. 4	51	OUT	IN	GND				GND							5.0 V	1A to 1Y	3	12	ns
		"	52	OUT	GND	IN				"							"	1B to 1Y	"	"	"
		"	53				OUT	IN	GND	"							"	2A to 2Y	"	"	"
		"	54				OUT	GND	IN	"							"	2B to 2Y	"	"	"
		"	55							"	IN	GND	OUT				"	3A to 3Y	"	"	"
		"	56							"	GND	IN	OUT				"	3B to 3Y	"	"	"
		"	57							"				IN	GND	OUT	"	4A to 4Y	"	"	"
		"	58							"				GND	IN	OUT	"	4B to 4Y	"	"	"
	t_{PHL}	"	59	OUT	IN	GND				"							"	1A to 1Y	"	10	"
		"	60	OUT	GND	IN				"							"	1B to 1Y	"	"	"
		"	61				OUT	IN	GND	"							"	2A to 2Y	"	"	"
		"	62				OUT	GND	IN	"							"	2B to 2Y	"	"	"
		"	63							"	IN	GND	OUT				"	3A to 3Y	"	"	"
		"	64							"	GND	IN	OUT				"	3B to 3Y	"	"	"
		"	65							"				IN	GND	OUT	"	4A to 4Y	"	"	"
		"	66							"				GND	IN	OUT	"	4B to 4Y	"	"	"
10	t_{PLH}	"	67-74	Same tests and terminal conditions as subgroup 9, except T_C = +125°C.															"	14	"
	t_{PHL}	"	75-82																"	11	"
11	Same tests, terminal conditions, and limits as subgroup 10, except T_C = -55°C.																				

1/ For cases X and Y, pins not referenced are N/C.

2/ I_{IL} limits shall be as follows:

Test	Min/Max limits in µA for circuit		
	A	B	C
I_{IL}	0/-100	0/-100	0/-100

3/ Method 3011 of MIL-STD-883 shall be used, except the output shall be as specified herein, and the output current shall be operating rather than short circuit current. The output conditions have been chosen to produce a current that closely approximates one half of the true short-circuit output current, I_{OS}.

4/ I_O limits for circuit C shall be -30 to -110.

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TABLE III. Group A inspection for device type 02.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases X, Y 1/	2	3	4	6	8	9	10	12	13	14	16	18	19	20	Measured terminal	Limits		Unit
			Cases A,B,C,D	1	2	3	4	5	6	7	8	9	10	11	12	13	14		Min	Max	
			Test no.	1A	1B	2A	2B	2C	2Y	GND	3Y	3A	3B	3C	1Y	1C	V_{CC}				
1 T_C = 25°C	V_{OH}	3006	1	0.8 V	0.8 V	0.9 V	0.8 V	0.8 V	-0.4 mA	GND	-0.4 mA	0.8 V	0.8 V	0.8 V	-0.4 mA	0.8 V	4.5 V	1Y	2.5		V
		"	2							"							"	2Y	"		"
		"	3							"							"	3Y	"		"
	V_{OL}	3007	4	2.0 V	GND					"					4.0 mA	GND	"	1Y		0.4	"
		"	5	GND	2.0 V					"					4.0 mA	GND	"	1Y		"	"
		"	6	GND	GND					"					4.0 mA	2.0 V	"	1Y		"	"
		"	7			2.0 V	GND	GND	4.0 mA	"							"	2Y		"	"
		"	8			GND	2.0 V	GND	4.0 mA	"							"	2Y		"	"
		"	9			GND	GND	2.0 V	4.0 mA	"							"	2Y		"	"
		"	10							"	4.0 mA	2.0 V	GND	GND			"	3Y		"	"
		"	11							"	4.0 mA	GND	2.0 V	GND			"	3Y		"	"
		"	12							"	4.0 mA	GND	GND	2.0 V			"	3Y		"	"
	V_{IC}		13	-18 mA						"							"	1A		-1.5	"
			14		-18 mA					"							"	1B		"	"
			15			-18 mA				"							"	1C		"	"
			16				-18 mA			"							"	2A		"	"
			17					-18 mA		"							"	2B		"	"
			18							"							"	2C		"	"
			19							"		-18 mA					"	3A		"	"
			20							"			-18 mA				"	3B		"	"
			21							"				-18 mA			"	3C		"	"
	I_{IL}	3009	22	0.4 V						"							5.5 V	1A	2/	2/	μA
		"	23		0.4 V					"							"	1B	"	"	"
		"	24			0.4 V				"							"	1C	"	"	"
		"	25				0.4 V			"							"	2A	"	"	"
		"	26					0.4 V		"							"	2B	"	"	"
		"	27							"							"	2C	"	"	"
		"	28							"		0.4 V					"	3A	"	"	"
		"	29							"			0.4 V				"	3B	"	"	"
		"	30							"				0.4 V			"	3C	"	"	"
	I_{IH1}	3010	31	2.7 V						"							"	1A		20	"
		"	32		2.7 V					"							"	1B		"	"
		"	33			2.7 V				"							"	1C		"	"
		"	34				2.7 V			"							"	2A		"	"
		"	35					2.7 V		"							"	2B		"	"
		"	36							"							"	2C		"	"
		"	37							"		2.7 V					"	3A		"	"
		"	38							"			2.7 V				"	3B		"	"
		"	39							"				2.7 V			"	3C		"	"
	I_{IH2}	"	40	7.0 V						"							"	1A		100	"
		"	41		7.0 V					"							"	1B		"	"
		"	42			7.0 V				"							"	1C		"	"
		"	43				7.0 V			"							"	2A		"	"
		"	44					7.0 V		"							"	2B		"	"
		"	45							"							"	2C		"	"
		"	46							"		7.0 V					"	3A		"	"
		"	47							"			7.0 V				"	3B		"	"
		"	48							"				7.0 V			"	3C		"	"
	I_O 3/	3011	49	GND	GND					"					2.25 V	GND	"	1Y	-30 4/	-112 4/	mA
		"	50			GND	GND	GND	2.25 V	"	2.25 V	GND	GND	GND			"	2Y	"	"	"
		"	51							"							"	3Y	"	"	"
	I_{CCH}	3005	52	GND	GND	GND	GND	GND		"		GND	GND	GND		GND	"	V_{CC}		1.8	"
	I_{CCL}	3005	53	4.5 V	4.5 V	4.5 V	4.5 V	4.5 V		"		4.5 V	4.5 V	4.5 V		4.5 V	"	V_{CC}		4.0	"

See footnotes at end of table.

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TABLE III. Group A inspection for device type 02 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases X, Y 1/	2	3	4	6	8	9	10	12	13	14	16	18	19	20	Measured terminal	Limits		Unit
			Cases A,B,C,D	1	2	3	4	5	6	7	8	9	10	11	12	13	14		Min	Max	
			Test no.	1A	1B	2A	2B	2C	2Y	GND	3Y	3A	3B	3C	1Y	1C	V_{CC}				
2	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = +125°C and V_{IC} tests are omitted.																				
3	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																				
9	t_{PLH}	3003	54	IN	GND					GND					OUT	GND	5.0 V	1A to 1Y	4	15	ns
T_C = 25°C		Fig. 4	55	GND	IN					"					OUT	GND	"	1B to 1Y	"	"	"
		"	56	GND	GND					"					OUT	IN	"	1C to 1Y	"	"	"
		"	57			IN	GND	GND	OUT	"							"	2A to 2Y	"	"	"
		"	58			GND	IN	GND	OUT	"							"	2B to 2Y	"	"	"
		"	59			GND	GND	IN	OUT	"							"	2C to 2Y	"	"	"
		"	60							"	OUT	IN	GND	GND			"	3A to 3Y	"	"	"
		"	61							"	OUT	GND	IN	GND			"	3B to 3Y	"	"	"
		"	62							"	OUT	GND	GND	IN			"	3C to 3Y	"	"	"
	t_{PHL}	"	63	IN	GND					"					OUT	GND	5.0 V	1A to 1Y	3	9	"
		"	64	GND	IN					"					OUT	GND	"	1B to 1Y	"	"	"
		"	65	GND	GND					"					OUT	IN	"	1C to 1Y	"	"	"
		"	66			IN	GND	GND	OUT	"							"	2A to 2Y	"	"	"
		"	67			GND	IN	GND	OUT	"							"	2B to 2Y	"	"	"
		"	68			GND	GND	IN	OUT	"							"	2C to 2Y	"	"	"
		"	69							"	OUT	IN	GND	GND			"	3A to 3Y	"	"	"
		"	70							"	OUT	GND	IN	GND			"	3B to 3Y	"	"	"
		"	71							"	OUT	GND	GND	IN			"	3C to 3Y	"	"	"
10	t_{PLH}	"	72-80	Same tests and terminal conditions as subgroup 9, except T_C = +125°C.															4	22	"
	t_{PHL}	"	82-87																3	10	"
11	Same tests, terminal conditions, and limits as subgroup 10, except T_C = -55°C.																				

1/ For cases X and Y, pins not referenced are N/C.

2/ I_{IL} limits shall be as follows:

Test	Min/Max limits in µA for circuit		
	A	B	C
I_{IL}	0/-100	0/-100	0/-100

3/ Method 3011 of MIL-STD-883 shall be used, except the output shall be as specified herein, and the output current shall be operating rather than short circuit current. The output conditions have been chosen to produce a current that closely approximates one half of the true short-circuit output current, I_{OS}.

4/ I_O limits for circuit C shall be -30 to -110 mA.

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6.3 Ordering data. The acquisition document should specify the following:

a. Complete part number (see 1.2).

b. Requirements for delivery of one copy of the quality conformance inspection data pertinent to the device inspection lot to be supplied with each shipment by the device manufacturer, if applicable.

c. Requirements for certificate of compliance, if applicable.

d. Requirements for notification of change of product or process to the contracting activity in addition to notification to the qualifying activity, if applicable.

e. Requirements for failure analysis (including required test condition of method 5003 of MIL-STD-883), corrective action, and reporting of results, if applicable.

f. Requirements for product assurance options.

g. Requirements for special carriers, lead lengths, or lead forming, if applicable. Unless otherwise specified, these requirements will not apply to direct purchase by or direct shipment to the Government.

h. Requirements for "JAN" marking.

6.4 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-M-38510, MIL-STD-1331, and as follows:

GND - - - - - - - - - - - - - - Ground zero voltage potential.

V_{IN} - - - - - - - - - - - - - - Voltage level at an input terminal.

I_{IN} - - - - - - - - - - - - - - Current flowing into an input terminal.

6.5 Logistic support. Lead materials and finishes (see 3.3) are interchangeable. Unless otherwise specified, microcircuits acquired for Government logistic support will be acquired to device class B (see 1.2.2), and lead finish C (see 3.3). Longer length leads and lead forming shall not affect the part number.

6.6 Substitutability. The cross-reference information below is presented for the convenience of users. Microcircuits covered by this specification will functionally replace the listed generic-industry type. Generic-industry microcircuit types may not have equivalent operational performance characteristics across military temperature ranges or reliability factors equivalent to MIL-M-38510 device types and may have slight physical variations in relation to case size. The presence of this information shall not be deemed as permitting substitution of generic-industry types for MIL-M-38510 types or as a waiver of any of the provisions of MIL-M-38510.

Military device type	Generic-industry type
01	54ALS02
02	54ALS27

6.7 Manufacturers' designation. Manufacturers' circuits which form a part of this specification are designated with an "X" as shown in table IV herein.

TABLE IV. Manufacturers' designation.

Device type	Circuits		
	A	B	C
	Texas Instruments	Motorola	National
01	X	X	X
02	X	X	X

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Air Force - 17

Review activities:
Air Force - 11, 19, 85, 99
DLA - ES

Agent:
DLA - ES

Preparing activity:
Air Force - 17

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a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

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